

Dnyaan Prasad Global University

Dr. D. Y. Patil Unitech Society — School of Technology and Research

EMMD: Question Bank on Unit-1 — Full Worked Solutions

Q1. Coefficient of Variation — Batsmen

Solution. Only the matches actually played are used.

Batsman-1 (played 4 matches): 56, 13, 39, 24.

$$\bar{x}_1 = \frac{56 + 13 + 39 + 24}{4} = \frac{132}{4} = 33$$

Deviations from mean: 23, -20, 6, -9. Squares: 529, 400, 36, 81, sum = 1046.

$$\sigma_1 = \sqrt{\frac{1046}{4}} = \sqrt{261.5} \approx 16.171$$

$$CV_1 = \frac{\sigma_1}{\bar{x}_1} \times 100 = \frac{16.171}{33} \times 100 \approx 49.00\%$$

Batsman-2 (played 7 matches): 20, 10, 5, 47, 29, 4, 12.

$$\bar{x}_2 = \frac{20 + 10 + 5 + 47 + 29 + 4 + 12}{7} = \frac{127}{7} \approx 18.143$$

Deviations: 1.857, -8.143, -13.143, 28.857, 10.857, -14.143, -6.143. Sum of squares ≈ 1431.71 .

$$\sigma_2 = \sqrt{\frac{1431.71}{7}} \approx \sqrt{204.53} \approx 14.297$$

$$CV_2 = \frac{14.297}{18.143} \times 100 \approx 78.80\%$$

Since $CV_1 < CV_2$, Batsman-1 has the lower relative dispersion.

Final Answer: $CV_1 \approx 49.00\%$, $CV_2 \approx 78.80\%$; **Batsman-1 is more consistent.**

Q2. Coefficient of Variation — Employees

Solution. Employee A: 18, 20, 22, 19, 21, 20, $n = 6$.

$$\bar{x}_A = \frac{18 + 20 + 22 + 19 + 21 + 20}{6} = \frac{120}{6} = 20$$

Deviations: -2, 0, 2, -1, 1, 0; squares: 4, 0, 4, 1, 1, 0, sum = 10.

$$\sigma_A = \sqrt{\frac{10}{6}} = \sqrt{1.667} \approx 1.291$$

$$CV_A = \frac{1.291}{20} \times 100 \approx 6.45\%$$

Employee B: 25, 12, 28, 15, 30, 10, $n = 6$.

$$\bar{x}_B = \frac{25 + 12 + 28 + 15 + 30 + 10}{6} = \frac{120}{6} = 20$$

Deviations: 5, -8, 8, -5, 10, -10; squares: 25, 64, 64, 25, 100, 100, sum = 378.

$$\sigma_B = \sqrt{\frac{378}{6}} = \sqrt{63} \approx 7.937$$

$$CV_B = \frac{7.937}{20} \times 100 \approx 39.69\%$$

Since $CV_A \ll CV_B$, Employee A's daily output is far more stable.

Final Answer: $CV_A \approx 6.45\%$, $CV_B \approx 39.69\%$; **Employee A is more consistent.**

Q3. Standard Deviation of First n Natural Numbers

Solution. Let the values be $1, 2, \dots, n$. Their mean is

$$\bar{x} = \frac{1 + 2 + \dots + n}{n} = \frac{1}{n} \cdot \frac{n(n+1)}{2} = \frac{n+1}{2}.$$

The variance (about the mean) equals the second raw moment minus the square of the mean:

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n i^2 - \bar{x}^2.$$

Using $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$:

$$\sigma^2 = \frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2}\right)^2 = (n+1) \left[\frac{2n+1}{6} - \frac{n+1}{4} \right].$$

Common denominator 12:

$$\sigma^2 = (n+1) \cdot \frac{2(2n+1) - 3(n+1)}{12} = (n+1) \cdot \frac{4n+2-3n-3}{12} = (n+1) \cdot \frac{n-1}{12} = \frac{n^2-1}{12}.$$

Hence

$$\sigma = \sqrt{\frac{n^2-1}{12}}. \quad \blacksquare$$

Final Answer: $\sigma = \sqrt{\frac{n^2-1}{12}}$ — proved.

Q4. Mean, Variance, SD, CV of Quiz Marks

Solution. $N = \sum f = 2 + 4 + 7 + 10 + 9 + 6 + 2 = 40$. Use the step-deviation method with assumed mean $A = 14$, $h = 2$, $d = \frac{x-A}{h}$.

x	f	$d = (x - 14)/2$	fd	fd^2
8	2	-3	-6	18
10	4	-2	-8	16
12	7	-1	-7	7
14	10	0	0	0
16	9	1	9	9
18	6	2	12	24
20	2	3	6	18
Σ	40		6	92

Mean:

$$\bar{x} = A + h \cdot \frac{\sum fd}{N} = 14 + 2 \cdot \frac{6}{40} = 14 + 0.3 = 14.3$$

Variance:

$$\sigma^2 = h^2 \left[\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N} \right)^2 \right] = 4 \left[\frac{92}{40} - \left(\frac{6}{40} \right)^2 \right] = 4[2.3 - 0.0225] = 4(2.2775) = 9.11$$

Standard deviation: $\sigma = \sqrt{9.11} \approx 3.0183$ **Coefficient of variation:** $CV = \frac{\sigma}{\bar{x}} \times 100 = \frac{3.0183}{14.3} \times 100 \approx 21.11\%$
Final Answer: $\bar{x} = 14.3$, $\sigma^2 = 9.11$, $\sigma \approx 3.0183$, $CV \approx 21.11\%$.

Q5. Moments and Shape of a Symmetric Distribution

Solution. $x : 1, 2, 3, 4, 5$, $f : 1, 3, 4, 3, 1$, $N = 12$.

x	f	fx	fx^2	fx^3	fx^4
1	1	1	1	1	1
2	3	6	12	24	48
3	4	12	36	108	324
4	3	12	48	192	768
5	1	5	25	125	625
Σ	12	36	122	450	1766

(i) Raw moments about origin:

$$\mu'_1 = \frac{36}{12} = 3, \quad \mu'_2 = \frac{122}{12} = \frac{61}{6}, \quad \mu'_3 = \frac{450}{12} = \frac{75}{2}, \quad \mu'_4 = \frac{1766}{12} = \frac{883}{6}$$

(ii) Central moments using $\mu_1 = 0$, $\mu_2 = \mu'_2 - \mu_1'^2$, $\mu_3 = \mu'_3 - 3\mu'_1\mu'_2 + 2\mu_1'^3$, $\mu_4 = \mu'_4 - 4\mu'_1\mu'_3 + 6\mu_1'^2\mu'_2 - 3\mu_1'^4$:

$$\mu_2 = \frac{61}{6} - 9 = \frac{7}{6}, \quad \mu_3 = \frac{75}{2} - 3(3)\frac{61}{6} + 2(27) = \frac{75}{2} - \frac{549}{6} + 54 = 0$$

$$\mu_4 = \frac{883}{6} - 4(3)\frac{75}{2} + 6(9)\frac{61}{6} - 3(81) = \frac{883}{6} - 450 + 549 - 243 = \frac{19}{6}$$

(iii) **Moment coefficients:**

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}} = 0, \quad \beta_1 = \frac{\mu_3^2}{\mu_2^3} = 0$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{19/6}{(7/6)^2} = \frac{19/6}{49/36} = \frac{19 \times 36}{6 \times 49} = \frac{114}{49} \approx 2.327$$

$$\gamma_2 = \beta_2 - 3 = \frac{114}{49} - 3 = -\frac{33}{49}$$

(iv) **Interpretation:** $\mu_3 = 0 \Rightarrow$ the distribution is symmetric; $\beta_2 < 3$ ($\gamma_2 < 0$) $\Rightarrow$ platykurtic (flatter than normal).

Final Answer: $\mu'_1 = 3$, $\mu'_2 = \frac{61}{6}$, $\mu'_3 = \frac{75}{2}$, $\mu'_4 = \frac{883}{6}$; $\mu_2 = \frac{7}{6}$, $\mu_3 = 0$, $\mu_4 = \frac{19}{6}$; $\gamma_1 = 0$, $\beta_1 = 0$, $\beta_2 = \frac{114}{49}$, $\gamma_2 = -\frac{33}{49}$ — symmetric, platykurtic.

Q6. Moments and Shape — Second Distribution

Solution. $x : 0, 1, 2, 3, 4$, $f : 1, 2, 4, 2, 1$, $N = 10$.

x	f	fx	fx^2	fx^3	fx^4
0	1	0	0	0	0
1	2	2	2	2	2
2	4	8	16	32	64
3	2	6	18	54	162
4	1	4	16	64	256
Σ	10	20	52	152	484

Raw moments:

$$\mu'_1 = \frac{20}{10} = 2, \quad \mu'_2 = \frac{52}{10} = \frac{26}{5}, \quad \mu'_3 = \frac{152}{10} = \frac{76}{5}, \quad \mu'_4 = \frac{484}{10} = \frac{242}{5}$$

Central moments:

$$\mu_2 = \frac{26}{5} - 4 = \frac{6}{5}, \quad \mu_3 = \frac{76}{5} - 3(2)\frac{26}{5} + 2(8) = \frac{76}{5} - \frac{156}{5} + 16 = 0$$

$$\mu_4 = \frac{242}{5} - 4(2)\frac{76}{5} + 6(4)\frac{26}{5} - 3(16) = \frac{242}{5} - \frac{608}{5} + \frac{624}{5} - 48 = \frac{18}{5}$$

Skewness: $\gamma_1 = 0$, $\beta_1 = \mu_3^2/\mu_2^3 = 0$ (distribution is symmetric).

Kurtosis:

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{18/5}{(6/5)^2} = \frac{18/5}{36/25} = \frac{18 \times 25}{5 \times 36} = \frac{5}{2}, \quad \gamma_2 = \beta_2 - 3 = -\frac{1}{2}$$

Since $\beta_2 < 3$, the distribution is platykurtic.

Final Answer: $\mu'_1 = 2$, $\mu'_2 = \frac{26}{5}$, $\mu'_3 = \frac{76}{5}$, $\mu'_4 = \frac{242}{5}$; $\mu_2 = \frac{6}{5}$, $\mu_3 = 0$, $\mu_4 = \frac{18}{5}$; $\gamma_1 = 0$, $\beta_1 = 0$, $\beta_2 = \frac{5}{2}$, $\gamma_2 = -\frac{1}{2}$ — symmetric, platykurtic.

Q7. Moments About $x = 4$ — Skewness and Kurtosis**Solution.** Given $\mu'_1 = 2$, $\mu'_2 = 20$, $\mu'_3 = 40$, $\mu'_4 = 400$ (moments about $A = 4$).(i) **Central moments** (using $\mu_1 = 0$ always, and the standard reduction formulas):

$$\mu_2 = \mu'_2 - \mu_1'^2 = 20 - 4 = 16$$

$$\mu_3 = \mu'_3 - 3\mu'_1\mu'_2 + 2\mu_1'^3 = 40 - 3(2)(20) + 2(8) = 40 - 120 + 16 = -64$$

$$\mu_4 = \mu'_4 - 4\mu'_1\mu'_3 + 6\mu_1'^2\mu'_2 - 3\mu_1'^4 = 400 - 4(2)(40) + 6(4)(20) - 3(16) = 400 - 320 + 480 - 48 = 512$$

(ii) **Skewness and kurtosis:**

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}} = \frac{-64}{16^{3/2}} = \frac{-64}{64} = -1, \quad \beta_1 = \gamma_1^2 = 1$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{512}{256} = 2, \quad \gamma_2 = \beta_2 - 3 = -1$$

(iii) **Mean and SD:**

$$\bar{x} = A + \mu'_1 = 4 + 2 = 6, \quad \sigma = \sqrt{\mu_2} = \sqrt{16} = 4$$

(iv) **Comment:** $\gamma_1 = -1 < 0 \Rightarrow$ negatively skewed; $\beta_2 = 2 < 3 \Rightarrow$ platykurtic.

Final Answer: $\mu_2 = 16$, $\mu_3 = -64$, $\mu_4 = 512$; $\gamma_1 = -1$, $\beta_1 = 1$, $\beta_2 = 2$, $\gamma_2 = -1$; $\bar{x} = 6$, $\sigma = 4$ — negatively skewed, platykurtic.

Q8. Moments About $x = 2$ — Skewness and Kurtosis**Solution.** Given $\mu'_1 = 1$, $\mu'_2 = 4$, $\mu'_3 = 16$, $\mu'_4 = 64$ (moments about $A = 2$).(i) **Central moments:**

$$\mu_2 = 4 - 1^2 = 3$$

$$\mu_3 = 16 - 3(1)(4) + 2(1)^3 = 16 - 12 + 2 = 6$$

$$\mu_4 = 64 - 4(1)(16) + 6(1)^2(4) - 3(1)^4 = 64 - 64 + 24 - 3 = 21$$

(ii) **Skewness and kurtosis:**

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}} = \frac{6}{3\sqrt{3}} = \frac{2}{\sqrt{3}} = \sqrt{\frac{4}{3}} \approx 1.1547, \quad \beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{36}{27} = \frac{4}{3}$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{21}{9} = \frac{7}{3}, \quad \gamma_2 = \beta_2 - 3 = -\frac{2}{3}$$

(iii) **Mean and SD:**

$$\bar{x} = A + \mu'_1 = 2 + 1 = 3, \quad \sigma = \sqrt{\mu_2} = \sqrt{3} \approx 1.7321$$

(iv) **Comment:** $\gamma_1 > 0 \Rightarrow$ positively skewed; $\beta_2 < 3 \Rightarrow$ platykurtic.

Final Answer: $\mu_2 = 3$, $\mu_3 = 6$, $\mu_4 = 21$; $\gamma_1 = \frac{2}{\sqrt{3}} \approx 1.1547$, $\beta_1 = \frac{4}{3}$, $\beta_2 = \frac{7}{3}$, $\gamma_2 = -\frac{2}{3}$; $\bar{x} = 3$, $\sigma = \sqrt{3}$ — positively skewed, platykurtic.

Q9. Karl Pearson's Coefficient of Skewness (Median Form)**Solution.** Data: 4, 5, 6, 7, 8, 9, 17, $n = 7$.

$$\bar{x} = \frac{4 + 5 + 6 + 7 + 8 + 9 + 17}{7} = \frac{56}{7} = 8$$

Arranged data is already sorted; the middle (4th) value is the median: Median = 7.

Deviations from mean: -4, -3, -2, -1, 0, 1, 9; squares: 16, 9, 4, 1, 0, 1, 81, sum = 112.

$$\sigma = \sqrt{\frac{112}{7}} = \sqrt{16} = 4$$

Since the median form of Pearson's coefficient is $Sk = \frac{3(\bar{x} - \text{Median})}{\sigma}$:

$$Sk = \frac{3(8 - 7)}{4} = \frac{3}{4} = 0.75$$

As $Sk > 0$, the distribution is positively skewed (the long tail is on the right, caused by the outlier value 17).**Final Answer:** $\bar{x} = 8$, Median = 7, $\sigma = 4$, $Sk = 0.75$ — positively skewed.**Q10. Karl Pearson's Coefficient of Skewness (Mode Form)****Solution.** Data: 4, 5, 5, 5, 6, 7, 8, 9, 14, $n = 9$.

$$\bar{x} = \frac{4 + 5 + 5 + 5 + 6 + 7 + 8 + 9 + 14}{9} = \frac{63}{9} = 7$$

The value 5 occurs most often (three times), so Mode = 5.

Deviations from mean: -3, -2, -2, -2, -1, 0, 1, 2, 7; squares: 9, 4, 4, 4, 1, 0, 1, 4, 49, sum = 76.

$$\sigma = \sqrt{\frac{76}{9}} = \frac{\sqrt{76}}{3} \approx 2.9059$$

Using the mode form $Sk = \frac{\bar{x} - \text{Mode}}{\sigma}$:

$$Sk = \frac{7 - 5}{\sqrt{76}/3} = \frac{2 \times 3}{\sqrt{76}} = \frac{6}{\sqrt{76}} = \frac{3}{\sqrt{19}} \approx 0.6882$$

The positive sign shows the distribution has a longer tail on the right (positive skewness), consistent with the high value 14 pulling the mean above the mode.

Final Answer: $\bar{x} = 7$, Mode = 5, $\sigma = \sqrt{76}/3 \approx 2.9059$, $Sk = 3/\sqrt{19} \approx 0.6882$ — positive skewness.**Q11. Karl Pearson Correlation Coefficient (Set 1)****Solution.** $X : -1, 1, 2, 4, 6, 7$; $Y : -1, 2, 3, 3, 5, 8$; $n = 6$.

$$\bar{X} = \frac{-1 + 1 + 2 + 4 + 6 + 7}{6} = \frac{19}{6}, \quad \bar{Y} = \frac{-1 + 2 + 3 + 3 + 5 + 8}{6} = \frac{20}{6} = \frac{10}{3}$$

X	Y	$x = X - \bar{X}$	$y = Y - \bar{Y}$	x^2	y^2	xy
-1	-1	-25/6	-13/3	625/36	169/9	325/18
1	2	-13/6	-4/3	169/36	16/9	52/18
2	3	-7/6	-1/3	49/36	1/9	7/18
4	3	5/6	-1/3	25/36	1/9	-5/18
6	5	17/6	5/3	289/36	25/9	85/18
7	8	23/6	14/3	529/36	196/9	322/18
Totals:				281/6	136/3	131/3

$$r = \frac{\sum xy}{\sqrt{\sum x^2 \cdot \sum y^2}} = \frac{131/3}{\sqrt{(281/6)(136/3)}} = \frac{131}{\sqrt{19108}} \approx 0.9477$$

Since r is close to 1, X and Y show a strong positive (linear) correlation.

Final Answer: $r = \frac{131}{\sqrt{19108}} \approx 0.9477$ — strong positive correlation.

Q12. Ad Spend vs. Downloads

Solution. X (spend, thousand rupees): 2, 4, 6, 8, 10, 12; Y (downloads): 30, 50, 65, 80, 95, 110; $n = 6$.

$$\bar{X} = \frac{2 + 4 + 6 + 8 + 10 + 12}{6} = 7, \quad \bar{Y} = \frac{30 + 50 + 65 + 80 + 95 + 110}{6} = \frac{430}{6} = \frac{215}{3}$$

X	Y	$x = X - 7$	$y = Y - \frac{215}{3}$	x^2	y^2	xy
2	30	-5	-125/3	25	15625/9	625/3
4	50	-3	-65/3	9	4225/9	195/3
6	65	-1	-20/3	1	400/9	20/3
8	80	1	25/3	1	625/9	25/3
10	95	3	70/3	9	4900/9	210/3
12	110	5	115/3	25	13225/9	575/3
Totals:				70	13000/3	550

$$r = \frac{\sum xy}{\sqrt{\sum x^2 \cdot \sum y^2}} = \frac{550}{\sqrt{70 \times 13000/3}} \approx \frac{550}{550.76} \approx 0.9986$$

This is an extremely strong positive correlation: downloads rise almost proportionally with ad spend.

Final Answer: $r \approx 0.9986$ — extremely strong positive correlation between ad spend and downloads.

Q13. Practice Hours vs. Test Scores

Solution. X (hours): 2, 3, 4, 5, 6, 7; Y (scores): 40, 45, 52, 60, 65, 72; $n = 6$.

$$\bar{X} = \frac{2 + 3 + 4 + 5 + 6 + 7}{6} = \frac{27}{6} = 4.5, \quad \bar{Y} = \frac{40 + 45 + 52 + 60 + 65 + 72}{6} = \frac{334}{6} = \frac{167}{3}$$

X	Y	$x = X - 4.5$	$y = Y - \frac{167}{3}$	x^2	y^2	xy
2	40	-2.5	-47/3	6.25		
3	45	-1.5	-32/3	2.25		
4	52	-0.5	-11/3	0.25		
5	60	0.5	13/3	0.25		
6	65	1.5	28/3	2.25		
7	72	2.5	49/3	6.25		
Totals:				17.5	2236/3	114

$$r = \frac{\sum xy}{\sqrt{\sum x^2 \cdot \sum y^2}} = \frac{114}{\sqrt{17.5 \times 2236/3}} \approx \frac{114}{114.21} \approx 0.9982$$

Practice hours and test scores are extremely strongly positively correlated.

Final Answer: $r \approx 0.9982$ — extremely strong positive correlation between practice hours and test scores.

Q14. Means and Correlation from Regression Lines (Set 1)

Solution. Given regression lines:

$$3x + 2y - 26 = 0 \quad (1) \qquad 6x + y - 31 = 0 \quad (2)$$

Both regression lines pass through the point $(\bar{x}, \bar{y})$, so we solve them simultaneously.

From (2): $y = 31 - 6x$. Substituting into (1):

$$3x + 2(31 - 6x) - 26 = 0 \Rightarrow 3x + 62 - 12x - 26 = 0 \Rightarrow -9x + 36 = 0 \Rightarrow x = 4$$

Then $y = 31 - 6(4) = 31 - 24 = 7$.

$$\bar{x} = 4, \quad \bar{y} = 7$$

(ii) Correlation coefficient: Write each line in slope form to identify which is y on x and which is x on y .

Treat (1) as regression of y on x : $2y = -3x + 26 \Rightarrow y = -\frac{3}{2}x + 13$, so $b_{yx} = -\frac{3}{2}$.

Treat (2) as regression of x on y : $6x = -y + 31 \Rightarrow x = -\frac{1}{6}y + \frac{31}{6}$, so $b_{xy} = -\frac{1}{6}$.

$$r^2 = b_{yx} \cdot b_{xy} = \left(-\frac{3}{2}\right) \left(-\frac{1}{6}\right) = \frac{1}{4} \Rightarrow r = \pm \frac{1}{2}$$

Since both regression coefficients are negative, r must also be negative:

$$r = -\frac{1}{2}$$

Final Answer: $\bar{x} = 4, \bar{y} = 7; r = -\frac{1}{2}$.

Q15. Means and Correlation from Regression Lines (Set 2)**Solution.** Given regression lines:

$$8x - 10y + 66 = 0 \quad (1) \qquad 14x - 18y + 124 = 0 \quad (2)$$

Solving simultaneously for $(\bar{x}, \bar{y})$: multiply (1) by 7 and (2) by 4:

$$56x - 70y + 462 = 0, \qquad 56x - 72y + 496 = 0$$

Subtracting: $2y - 34 = 0 \Rightarrow y = 17$. Substitute into (1): $8x - 170 + 66 = 0 \Rightarrow 8x = 104 \Rightarrow x = 13$.

$$\boxed{\bar{x} = 13, \quad \bar{y} = 17}$$

(ii) Correlation coefficient: Take (1) as y on x : $10y = 8x + 66 \Rightarrow y = 0.8x + 6.6$, so $b_{yx} = 0.8 = \frac{4}{5}$.

Take (2) as x on y : $14x = 18y - 124 \Rightarrow x = \frac{18}{14}y - \frac{124}{14} = \frac{9}{7}y - \frac{62}{7}$, so $b_{xy} = \frac{9}{7}$.

$$r^2 = b_{yx} \cdot b_{xy} = \frac{4}{5} \cdot \frac{9}{7} = \frac{36}{35}$$

Since r^2 must not exceed 1, the assignment must be swapped: take (2) as y on x and (1) as x on y instead.

From (2) as y on x : $18y = 14x + 124 \Rightarrow y = \frac{7}{9}x + \frac{62}{9}$, so $b_{yx} = \frac{7}{9}$.

From (1) as x on y : $8x = 10y - 66 \Rightarrow x = \frac{10}{8}y - \frac{66}{8} = \frac{5}{4}y - \frac{33}{4}$, so $b_{xy} = \frac{5}{4}$.

$$r^2 = \frac{7}{9} \cdot \frac{5}{4} = \frac{35}{36} \Rightarrow r = \sqrt{\frac{35}{36}} = \frac{\sqrt{35}}{6} \approx 0.9860$$

Both regression coefficients are positive, so r is positive.

$$\boxed{\text{Final Answer: } \bar{x} = 13, \bar{y} = 17; r = \frac{\sqrt{35}}{6} \approx 0.9860.}$$

Q16. Regression Line of y on x **Solution.** $x : 1, 2, 3, 4, 5$; $y : 2, 4, 5, 4, 6$; $n = 5$.

$$\bar{x} = \frac{1 + 2 + 3 + 4 + 5}{5} = 3, \qquad \bar{y} = \frac{2 + 4 + 5 + 4 + 6}{5} = \frac{21}{5} = 4.2$$

x $(x - \bar{x})(y - \bar{y})$	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$
1	2	-2	-2.2	4
4.4				
2	4	-1	-0.2	1
0.2				
3	5	0	0.8	0
0				
4	4	1	-0.2	1
-0.2				
5	6	2	1.8	4
3.6				
Totals:				10
8				

$$b_{yx} = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sum(x - \bar{x})^2} = \frac{8}{10} = 0.8$$

Regression line: $y - \bar{y} = b_{yx}(x - \bar{x}) \Rightarrow y - 4.2 = 0.8(x - 3) \Rightarrow y = 0.8x + 1.8$.

At $x = 6$: $\hat{y} = 0.8(6) + 1.8 = 4.8 + 1.8 = 6.6$.

Final Answer: Regression line: $y = 1.8 + 0.8x$; at $x = 6$, $\hat{y} = 6.6$.

Q17. Two Regression Lines and Estimation

Solution. $x : 6, 2, 10, 4, 8$; $y : 9, 11, 5, 8, 7$; $n = 5$.

$$\bar{x} = \frac{6 + 2 + 10 + 4 + 8}{5} = 6, \quad \bar{y} = \frac{9 + 11 + 5 + 8 + 7}{5} = 8$$

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(y - \bar{y})^2$	$(x - \bar{x})(y - \bar{y})$
6	9	0	1	0	1	0
2	11	-4	3	16	9	-12
10	5	4	-3	16	9	-12
4	8	-2	0	4	0	0
8	7	2	-1	4	1	-2
Totals:				40	20	-26

$$b_{yx} = \frac{-26}{40} = -\frac{13}{20}, \quad b_{xy} = \frac{-26}{20} = -\frac{13}{10}$$

Regression line of y on x :

$$y - \bar{y} = b_{yx}(x - \bar{x}) \Rightarrow y - 8 = -\frac{13}{20}(x - 6)$$

Regression line of x on y :

$$x - \bar{x} = b_{xy}(y - \bar{y}) \Rightarrow x - 6 = -\frac{13}{10}(y - 8)$$

(i) Estimate y at $x = 12$:

$$y - 8 = -\frac{13}{20}(12 - 6) = -\frac{13}{20}(6) = -3.9 \Rightarrow y = 8 - 3.9 = 4.1$$

(ii) Estimate x at $y = 8$:

$$x - 6 = -\frac{13}{10}(8 - 8) = 0 \Rightarrow x = 6$$

Final Answer: Regression lines: $y - 8 = -\frac{13}{20}(x - 6)$ and $x - 6 = -\frac{13}{10}(y - 8)$; (i) $\hat{y} = 4.1$ at $x = 12$; (ii) $\hat{x} = 6$ at $y = 8$.